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| **Fiscal Year:** | FY 2005 | **Task Last Updated:** | FY 01/03/2005 |
| **PI Name:** | Larson, Ronald G Ph.D. | | |
| **Project Title:** | Microfluidic and Dielectric Processing of DNA | | |
| **Division Name:** | Physical Sciences | | |
| **Program/Discipline:** | FLUID PHYSICS | | |
| **Program/Discipline--Element/Subdiscipline:** | FLUID PHYSICS--Cell science | | |
| **Joint Agency Name:** | | **TechPort:** | No |
| **Human Research Program Elements:** | None | | |
| **Human Research Program Risks:** | None | | |
| **Space Biology Element:** | None | | |
| **Space Biology Cross-Element Discipline:** | None | | |
| **Space Biology Special Category:** | None | | |
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| **Zip Code:** | 48109-2125 | **Congressional District:** | 15 |
| **Comments:** | | | |
| **Project Type:** | INTERDISCIPLINARY (2004-2007) | **Solicitation / Funding Source:** | 00-HEDS-02 |
| **Start Date:** | 02/13/2002 | **End Date:** | 02/04/2006 |
| **No. of Post Docs:** | 1 | **No. of PhD Degrees:** | 0 |
| **No. of PhD Candidates:** | 0 | **No. of Master' Degrees:** | 0 |
| **No. of Master's Candidates:** | 1 | **No. of Bachelor's Degrees:** | 0 |
| **No. of Bachelor's Candidates:** | 0 | **Monitoring Center:** | NASA GRC |
| **Contact Monitor:** | Griffin, DeVon | **Contact Phone:** | 216-433-8109 |
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| **Flight Program:** | | | |
| **Flight Assignment:** | | | |
| **Key Personnel Changes/Previous PI:** | | | |
| **COI Name (Institution):** | | | |
| **Grant/Contract No.:** | NAG 3 2708 | | |
| **Performance Goal No.:** | | | |
| **Performance Goal Text:** | | | |

Ground-based research to develop the science and technology of microfluidic and dielectric movement and attachment of DNA onto surfaces for subsequent DNA processing and analysis. This will follow up on NASA grant NAG3-2134 by using the flow fields studied in that grant to stretch and deposit double stranded DNA onto treated substrates, and to study the hydrodynamic and physico-chemical forces controlling the stretching and deposition processes. The flows to be used are: (1) a simple shearing flow produced by a rotating parallel disk parallel to a stationary disk, and (2) the flow in an evaporating droplet. In addition to these, fundamental studies will be carried out on dielectrically-driven stretching and movement of DNA molecules between electrodes. We will study DNA flow and deposition in (1) a simple shearing flow produced by a rotating parallel disk parallel to a stationary disk, and (2) the flow in an evaporating droplet. Finally, (3) we will determine the mechanism of dielectrically-driven stretching and movement of DNA molecules between electrodes.
Using a high-resolution CCD camera, we will measure quantitatively the stretching and anchoring of DNA on substrates

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| **Task Description:** | in these two flows. We will develop a model for the stretching of DNA near surfaces and for the adhesive forces exerted by the surfaces on the DNA. By comparing detailed distributions of molecular stretch of DNA adsorbed onto the surfaces with Brownian dynamics simulations of these distributions, we will validate the DNA deposition and adhesion model. We will also explore the use of confocal microscopy for quantitative measurement of the droplet-drying flow field. Stretching and anchoring DNA molecules is an important first step to studying DNA/protein interactions, including repair of double-strand breaks caused by ionizing radiation in space. The deposition of DNA using micro-flows is also interest in genomics and micro-arrays and so has wide applications. The addition of electric field effects provides an alternative means of stretching DNA molecules while retaining them in solution for restriction digestion and other applications. Stretching and anchoring DNA molecules is an important first step to studying DNA/protein interactions, including repair of double-strand breaks caused by ionizing radiation in space. |
| **Rationale for HRP Directed Research:** | |
| **Research Impact/Earth Benefits:** | Advanced methods of manipulating, stretching, aligning, and depositing DNA molecules will further study of DNA interactions with proteins, which is key to packaging, transcription, and repair of DNA, all vital processes with close connections to disease states, including cancer. |
| **Task Progress:** | This year we completed study of the flow in a drying octane droplet, whose surface remains clean for much longer periods than does water. To study this flow conveniently, we have developed an imaging method using a micro laser sheet, and a method of synthesizing fluorescent particles that are stable in octane, and of sufficiently large size to be viewed as flow tracers in the laser sheet. The synthesis of these polymethyl methacrylate (PMMA) particles constitutes an unexpected breakthrough that transcended the original goals of the project. Using these particles, we successfully imaged a strong Marangoni flow in a sessile droplet, for the first time. In addition, to correct for the effects of refraction of fluorescent light as it emerges from the droplet, we developed a ray-tracing algorithm, that allows us to compare quantitatively the predicted, with the observed, particle recirculation patterns in the micro-flow in the droplet. We have also observed DNA stretch at and near surfaces in simple shearing flows. We initially expected DNA to stretch DNA better near surfaces in simple shearing flow than in the droplet-drying flow, but found poorer stretch, evidently due to a hydrodynamic interactions, whereby stretched molecules are repulsed from the wall. To counter this effect, in preliminary work, we increased the DNA concentration to screen out the hydrodynamic interactions, and found that the DNA became much more concentrated at the wall than in very dilute solutions. |
| **Bibliography Type:** | Description: (Last Updated: 02/25/2006) |
| **Abstracts for Journals and Proceedings** | 1. Hu, H.; Larson, R.G "Preparation of fluorescent particles with long excitation and emission wavelengths in organic solvent" Langmuir<br>20 , Aug-2004 |
| **Abstracts for Journals and Proceedings** | H. Hu; R.G. Larson "One-Step Preparation of Highly Monodisperse Micron-Size Particles in Organic Solvents" Journal of the American Chemical Society<br>126 , Nov-2004 |